

Worksheet 1 · Spot on facts, pp. 226–227

Technologies that change society

The facts at a glance — your foundation for the Kolloquium and for any factual text on science and technology.

1 Developments in the 21st century

Genetic engineering

CRISPR = tool that *precisely edits* the genetic code of organisms.

- + drought-resistant crops; better treatments for incurable diseases (e.g. stem cells from human embryos)
- morality of the research; risks of human *interference* with DNA

Eco-technology

Ocean clean-ups try to remove the ever-growing amount of plastic waste.

- + solar, hydrogen and wind power are more affordable and efficient; the share of electric vehicles is rising
- fossil-fuel cars still outnumber e-cars in most countries; *raw materials* (lithium, rare-earth metals for batteries) create new environmental problems

AI & digital content

Large language models turn *prompts* into texts, audiovisual content and complex programs.

- + masses of information and data can be re-used in myriad contexts
- often impossible to know where content comes from and how trustworthy it is

2 Artificial intelligence (AI)

Definition: AI is the *simulation of human intelligence in machines* — learning, problem-solving, *reasoning* and perceiving. The field *encompasses* anything from online search engines to military weapons and is progressing rapidly.

Weak / narrow AI

single-task oriented, e.g. personal assistants, video games

Careful: most experts today use *general AI* (AGI) for a still hypothetical AI with human-level intelligence in *all* tasks and count self-driving cars as narrow AI.

→ **Key consequence:** checking data and recognising manipulation matter more than ever — for individuals, companies and governments.

Strong / general AI

more complex, human-like, e.g. (book) self-driving cars, robotic systems

Impact on society

- + benefits everyday life, education and health care
- + testing and scrutiny (e.g. combining research from diverse sources) can reduce *bias*, improve faulty systems and make slow processes far more efficient
- affects human employment
- ethical questions: telling human from AI texts, video *deepfakes*
- manipulation, misinformation, copyright infringements

3 Augmented reality (AR) and virtual reality (VR) — both types of extended reality (XR)

Augmented reality (AR)

What it does *superimposes* interactive digital elements (visual *overlays*, haptic feedback, sensory projections) on the real world → *alters* our perception of it

Devices devices with cameras (e.g. smartphone, tablet)

Used in entertainment, medicine, other commercial industries

Concerns data extracted by XR companies

Virtual reality (VR)

replaces our perception of the real world with a computer-generated 360° environment, fully *immersive* or non-immersive

headset and motion controllers; simulators

gaming, marketing, tourism, education, medicine, *warfare*

may affect a person's perception of reality; data extraction

Our texts: Virtual Helsinki = VR tourism (Baxter, p. 230) · mind uploading = a "virtual afterlife" (Graziano, p. 234)

4 Utopia or dystopia — where will technology take us?

Utopia Greek *ou* (not) + *topos* (place) = "no place"; sounds like *eu* (good) = "good place".

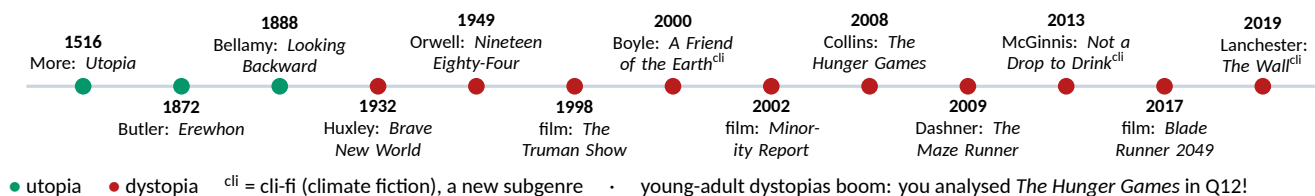
Coined by **Thomas More** in 1516: the perfect island of his book could never really exist.

→ an **imaginary, ideal society**; by highlighting aspects of a better world it *criticises contemporary society* and suggests changes.

Dystopia Greek *dys-* (bad) + *topos*.

Also political: a **critique of contemporary socio-political realities** that shows how bad things *could* become — worst-case scenarios if issues stay unaddressed.

→ often set in the **not-too-distant future**, in societies readers recognise; themes: politics, economics, identity, privacy, freedom, health, science and technology.



Worksheet 1 · Task

Kolloquium Carousel

In Part 2 of the Kolloquium you answer questions on topics from other semesters — no notes, no slides. Train it now.

How it works

- Form a double circle. **Inner circle = candidates, outer circle = examiners** (examiners hold this sheet).
- The examiner reads a card. The candidate thinks for 15 seconds, then answers for about **2 minutes**.
- The examiner asks the *follow-up question* (about 1 minute).
- The examiner scores with the rubric and gives **two stars and a wish** (1 minute).
- Outer circle moves one seat on. After two rounds, swap roles.

Answer like a candidate: D – E – E – E



Use the topic words: *to encompass, immersive, to alter, to differ from, interference, sustainable, contemporary, socio-political, issue, prediction*.
Link to what you know: our texts, *The Hunger Games*, the news.

Question cards (examiner: pick a card the candidate has not had yet — follow-up in italics)

1 AR vs VR

Explain the difference between augmented and virtual reality and give an example of each.

Follow-up: Which of the two will change everyday life more — and why?

2 AI

What is artificial intelligence? Distinguish between weak and strong AI.

Follow-up: Name one area where AI helps and one where it worries you.

3 Trust

Why is it becoming harder to know where online content comes from?

Follow-up: How can individuals protect themselves against manipulation?

4 Genes

Outline what CRISPR makes possible and why it is controversial.

Follow-up: Where would you draw the line — think of "designer babies".

5 Eco-tech

Is eco-technology the answer to our environmental problems?

Follow-up: What is the "small print" of electric cars?

6 Terms

Explain where the terms *utopia* and *dystopia* come from and what they mean.

Follow-up: Why might dystopias be more popular than utopias today?

7 Dystopia

Choose a dystopian novel or film you know. Which contemporary issue does it criticise?

Follow-up: Which technology today could lead to a similar society?

8 Big picture

Which technology on the page is the most life-changing?

Follow-up: How will it affect the lives of your generation's children?

Examiner's sheet 0 = not yet · 1 = partly · 2 = mostly · 3 = fully

	Round 1	Round 2
Candidate		
Content — facts and terms are accurate and precise	/3	/3
Structure — clear line of thought (D-E-E-E), gets to the point	/3	/3
Language — topic vocabulary, range, accuracy, fluency	/3	/3
Interaction — reacts to the follow-up spontaneously and to the point	/3	/3
Total	/12	/12
★ ★ _____	★ ★ _____	
wish _____	wish _____	

After the carousel — your Kolloquium note Which card was hardest for you? Note the facts, terms and one example you would need to answer it well next time.
